

Abhijith Senthilkumar

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Professional Summary

Master's student in Data Science at the University of Luxembourg with hands-on experience in data collection, cleaning, quality checks, validation, SQL reporting, and dashboard development. Built data quality gates, validation tests, and KPI dashboards across financial-services and credit-risk projects, with a consistent focus on accuracy, documentation, and reproducibility. Comfortable using Excel to inspect and explore raw data before moving into SQL and Python for deeper analysis. Collaborative team contributor with strong written and spoken English. Genuinely interested in applying data analytics within insurance, investment funds, and financial services, and keen to build domain knowledge in unit-linked insurance products.

Internship status: Mandatory curricular internship as part of the MSc Data Science programme; university internship agreement available.

Education

Master of Science in Data Science

2024 – present

University of Luxembourg

- **Relevant coursework:** Machine Learning (18/20), Data Visualization (17/20), Optimization and Numerical Probabilities (16/20), Probability and Statistics, Big Data Analytics, NoSQL Databases.

Bachelor in Computer Science Engineering

2018 – 2022

Anna University

- **Areas of focus:** Python, SQL, data analytics, data validation, and applied software development.

Technical Skills

Analytics & Reporting: Excel (data review and exploration), SQL, Power BI, Streamlit, KPI reporting, dashboarding, data cleaning, data validation, data quality checks, documentation

Databases & Programming: SQL, Python, R, DuckDB, MongoDB

Python Analytics Stack: Pandas, NumPy, scikit-learn, Matplotlib, Plotly, Jupyter Notebook

Financial & Business Analytics: Credit risk analytics, portfolio monitoring, forecasting, decision-support reporting, financial services

Tools & Practices: Git, Docker, unit testing of data pipelines, reproducible workflows, version-controlled reporting

Additional Technical Exposure: Machine learning (LightGBM, XGBoost, PyTorch, NLP) – used in academic research and personal projects

Languages: English (C2 – IELTS), French (A1)

Projects

Credit Risk BI & Decisioning Dashboard

Nov 2025 – Feb 2026

Technologies: Python, Streamlit, DuckDB, SQL, Excel (data review), scikit-learn, Docker

- Reviewed and explored raw applicant data in Excel before building a Python/SQL/Streamlit pipeline to clean, validate, and score credit risk data for portfolio reporting.
- Built automated data quality checks and unit-test-style validation rules to catch schema and consistency errors before they reached downstream reporting – [estimate: flagging roughly X% of injected/test errors during development].
- Designed SQL-based KPI reporting tables and versioned per-run CSV snapshots, enabling consistent, auditable reporting that supports business decision-making.
- Built an underwriting policy simulator translating model risk scores into **approve / review / decline** decisions under business-defined capacity and cost assumptions.

Walmart Weekly Sales Forecasting

Dec 2025 – Feb 2026

Technologies: Python, Pandas, NumPy, Matplotlib

- Cleaned and validated multi-year retail sales data, then built a forecasting workflow comparing department-level and aggregate-level demand planning approaches, evaluated using MAE, RMSE, and WAPE across **840 test points**.

Stocks Next-Day Returns Prediction & Portfolio News Sentiment Alerts

Oct 2025 – Mar 2026

Technologies: Python, scikit-learn, yfinance, NewsAPI

- Built leakage-aware data pipelines to clean, validate, and engineer features from market and news data, applying training-only preprocessing, chronological validation splits, and manual evaluation sets to keep results conservative and reproducible.

Experience

Research Assistant

Oct 2025 – Jan 2026

University of Luxembourg

- Collaborated with supervising faculty on **TimeCIM** and **TSGuard**, contributing to missing-data handling and structured evaluation for time-series research.
- Supported experiment tracking and documentation practices to keep research outputs reliable and reproducible across iterations.

- Communicated technical progress clearly to the research team, translating modelling work into interpretable findings.

ML Intern

Feb 2022 – May 2022

Toptech Solutions Private Limited, India

- Cleaned and validated cybersecurity datasets, preparing model-ready features for a network intrusion detection thesis project.
- Worked across data preprocessing, model development, and evaluation, building an early foundation in practical data analysis.

Research Intern / Visiting Student

Jul 2021 – Oct 2021

École de technologie supérieure

- Applied data-driven methods to network security and resource allocation problems within a structured research team, where reproducibility and clear technical communication were essential to team output.

Additional Analytics Experience

- Comfortable moving between technical implementation and business explanation: SQL tables, Excel data review, Python pipelines, dashboards, and written reports.
- Practical interest areas include financial services analytics, credit risk, investment data, reporting automation, and data quality – with a genuine interest in developing knowledge of unit-linked insurance and investment fund structures.

Certifications

Machine Learning Specialization – Stanford University / Coursera

[Certificate](#)

IBM Data Science Specialization – IBM / Coursera

[Certificate](#)